

Q20

Proof. Suppose for the contradiction, the curve $x^2 + y^2 + 3 = 0$ has rational points.

Then, there exist $x, y \in \mathbb{Q}$ that $x^2 + y^2 = 3$.

Since x and y are rational, we can write them with a common denominator:

$$x = \frac{a}{c}$$

$$y = \frac{b}{c}$$

where,

$$\exists a, b, c \in \mathbb{Z}, c \neq 0, \gcd(a, b, c) = 1.$$

Let us sub x and y into the curve,

$$\left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2 = 3$$

$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = 3$$

$$a^2 + b^2 = 3c^2$$

By lemma, we know under modulo 4, if n is odd, $n^2 \equiv 1 \pmod{4}$. If n is even, $n^2 \equiv 0 \pmod{4}$.

Case 1: c is odd

We get:

$$a^2 + b^2 \equiv 3 \pmod{4}$$

This is impossible to achieve.

Case 2: c is even

We get:

$$a^2 + b^2 \equiv 0 \pmod{4}$$

This is possible to achieve when a is even and b is even.

However, since $\gcd(a, b, c) = 1$, a, b, c cannot be all even.

Therefore, curve $x^2 + y^2 - 3 = 0$ does not have rational points.

□

Q21

Proof. Suppose, for contradiction, $x^2 + y^2 - 3^k = 0$ has rational solutions for k an odd, positive integer.

We get $k = 2n + 1, \exists n \in \mathbb{N}$, according to the definition of odd.

Let us sub $k = 2n + 1$ into $x^2 + y^2 - 3^k = 0$,

$$x^2 + y^2 = 3^{2n+1}$$

$$x^2 + y^2 = 3 \times 3^{2n}$$

$$x^2 + y^2 = 3 \times 9^n$$

$$a^2 + b^2 = 3 \cdot 9^n c^2 \pmod{4}$$

$$a^2 + b^2 = 3c^2 \pmod{4}$$

□

Q24

Proof. Suppose, for contradiction, JJJJ

□